

INSY7311

ASSIGNMENT 2-ST10257779

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1. The Introduction

Learners receive assistance from StudyMate, which serves as a mobile tool created for the management of academic completion dates and the organization of written notes. StudyMate answers a common difficult situation for individuals in education because academic information frequently stays distributed among different learning-management systems and digital calendars. When academic information exists in many places, the level of stress in individuals in education increases. Many people believe that StudyMate helps because StudyMate makes important academic actions observable and expected for the user. Through StudyMate, a central place for following tasks is provided to the user. StudyMate remains focused on usability so that learners do not miss their completion dates.

Earlier work in Assignment 1 showed the StudyMate abstract idea and pointed out the broad necessary needs of the users. This current Assignment 2 advances the project because known foundational rules of interaction-design are used for the updated first model. Each design decision was carefully chosen by the author to support learners during realistic study situations. It has been observed that the first answers required deeper engagement with academic concepts from the lectures. Because of this observation, the revised work explains why every choice is significant for the user. Clearer explanations are contained within this document to address previous evaluative remarks. StudyMate benefits from these stronger applications of foundational rules.

Three primary sections compose this document to show the development of StudyMate. In the first part, the evaluative remarks received from the previous submission are explained when the improvements are carried out. In the second part, the first model is expanded because five design rules are used, such as system status visibility and error prevention. Experts claim that consistency and standards are necessary for a high-quality digital environment. A strategic arrangement for assessment is presented in the third section. This strategic arrangement combines heuristic assessment with task-based testing of usability. An evidence-based method is followed by the academic paper to ensure StudyMate reaches a high academic level.

2. The Feedback that was received for Assignment 1 and How It Was Implemented

My answers for Assignment 1 said that they weren't based enough on the textbook and the class discussions. My lecturer also informed me that my descriptions were not detailed enough and I needed to use AI a bit more sparingly in order to copy and paste. This feedback showed that, although my Assignment 1 included relevant ideas about the Study Mate application, the report did not demonstrate enough independent academic engagement with the module content. Well, the task should have been improved with the help of class theory represented more clearly, with more details to the explanations and a more natural and detailed writing style.

There has been changes were made to Assignment 2 which responds to the feedback. First, the explanation of the prototype was made much longer. There is now a separate section for each design principle, instead of just a short mention of them. This makes it easier to see how the principles were

used in the Study Mate interface and how they make the app easier to use. Second, the report now has five different design principles, which is more in line with the assignment's requirements to include any five design principles or guidelines. The third, the evaluation plan has been fleshed out a lot more. The report now goes beyond just listing methods; it also explains the purpose of each method and why it should be used based on usability theory, and explains how each method will be used in real life. These changes make the report better academically and show that it follows the marking rubric more closely.

The report's structure and presentation are also better in this assignment. The content is better organised, the discussion goes more smoothly from design to evaluation, and the theory is more closely related to the prototype screens. This makes the report easier to read and shows that you know more about usability principles and evaluation methods.

3. The Expanded Low-Fidelity Prototype Using Five Design Principles

The low-fidelity prototype for Study Mate was first made to show off the app's main screens and navigation structure. The prototype for Assignment 1 already had a dashboard, a task management screen, a notes screen, and a navigation bar. These screens showed how students could see due dates, add tasks, organise notes, and switch between different parts of the app.

For Assignment 2, the prototype is improved by showing how five design principles make the user experience better overall. These principles are: Visibility, Feedback, Constraints, Mapping, and Conceptual Model. The Study Mate prototype uses each principle, (INTERACTION DESIGN, 2024).

3.1 Visibility

Visibility refers to making crucial decisions, activities, and data accessible such that users do not need to guess on the next step. According to Norman, a design must indicate to the users what they can do and where the key functions are to be found. It is particularly significant in the case of mobile apps, in which there is not a lot of space on the screen and individuals frequently have a short amount of time to complete certain tasks.

The home dashboard of the Study Mate prototype uses visibility. The most important data are shown on the dashboard at once, e.g. tasks and due dates today. The screen has an easy-to-identify Add Task button that enables students to add a new task or reminder. This makes the users easier to interact with the app since they can get a glimpse of what it is about without necessarily going through the app. The dashboard is a good place to begin then, in regards to getting things done and planning school. This is in line with what was said in the first Assignment 1 discussion, which said that the home screen and task visibility were important for quick use, (INTERACTION DESIGN, 2024).

3.2 Feedback

The Feedback is the exact system's response for an user's action. Any good feedback will let the users know if their actions were successful, unsuccessful or maybe still being processed. No feedback, the users might repeat actions, they'll become uncertain, or think the app isn't working properly, (INTERACTION DESIGN, 2024).

In this Study Mate original model, the feedback is only applied when the user adds the task. After the student inputs the task information and saves, the interface will show confirmation messages such as "The task has been saved successfully" and "The Reminder is set." This will assure the user that the information was processed and captured correctly. The use of feedback is very crucial in an

academic app, because the students need the confidence that deadlines and reminders have been stored correctly and accurately. This part was already introduced in Assignment 1 as part of the “Add Task” feature, where the model confirms successful user actions, (INTERACTION DESIGN, 2024).

The design principle helps improve the user experience because it reduces the doubt and it prevents all duplicate task entries. It also supports trust in the application. If students might feel confident that the app is responding correctly, they are more likely to continue using it way more consistently.

3.3 Constraints:

A wrong move slips less often when the path shapes behaviour. Blocking missteps happens by limiting choices ahead of time. Fewer errors pop up because options get narrowed early. Guiding hands come not from warnings after, but limits built in before. The setup steers people right by removing risky paths upfront.

A blank due date breaks the exact reminder functions in Study Mate’s task builder. Without that detail, deadlines can’t be tracked at all. The interface blocks saving until it’s filled - maybe by dimming the button or popping up “Due date required.” Missing fields were already flagged back in Assignment 1. Saving unfinished entries wasn’t allowed, (INTERACTION DESIGN, 2024).

3.4 Mapping:

What you see connects to what happens. Controls that are placed appropriately are intuitively sensible and meet the objectives of the user. Layouts are effective when movement or choice is in anticipated directions. When things are off, then it is usually because placement is in conflict with intuition rather than going with it, (INTERACTION DESIGN, 2024).

Inside the Study Mate test version, you see maps inside the Notes area. Each subject, like INSY6212 or INSY7311, holds its own set of notes - this feels familiar, since learners link topics to courses. Below those groups, or just next to them, sits the option to add a note; placement makes it obvious where entries go. Earlier details in Assignment 1 showed how organizing by course fits student thinking, which helps them grasp the system faster.

Because the app matches how students naturally sort their tasks, using it feels easier. Instead of learning a new way, they follow familiar patterns. The design cuts down mental strain by working like their own habits. Navigation becomes smoother when it acts like what they expect.

3.5 Conceptual Model:

Most people form mental pictures about how things operate - that’s their conceptual model. When an interface reflects a straightforward idea, actions lead to expected results. Because of this clarity, picking up usage patterns feels natural. Over days or weeks, recall stays strong without extra effort.

A fixed set of icons - Home, Tasks, Notes, Schedule, Search - anchors the bottom of key screens in the Study Mate prototype. Since these symbols do not shift position during use, users begin to link each one with a core function. Stability in layout comes from this unchanging placement, offering clarity over time. Such consistency has been mentioned earlier work as reducing mental load, and making transitions feel natural. Since the visual cues do not move, learners do not use much energy trying to follow where to go.

Because students open learning apps so many times, it matters how things are set up. When screens follow a pattern, picking up new parts gets easier. Jumping between sections feels smoother if layout stays consistent. Learning sticks better when navigation makes sense every time. Familiar design means less confusion, more doing, (INTERACTION DESIGN, 2024).

4. Evaluation Plan for the User Interface Prototype:

A review plan needs to be developed to show whether the Study Mate interface is effective, efficient and satisfying to use. In usability design, testing is needed so that a prototype is not merely a visualised organised material, but one which is practical to the actual users. Because Study Mate is supposed to assist students who have deadlines, manage the tasks and organize their notes, it is necessary to test whether the design in question can help the user to complete the tasks in question with ease.

The evaluation plan suggested to be used in Study Mate consists of two approaches: heuristic evaluation and task based usability testing. These two are complementary approaches. In heuristic evaluation, specialists or reviewers test the interface with principles of known usability concepts, whereas task-based usability testing is done by real users in performing realistic tasks with the prototype. Both approaches will give a more comprehensive idea of the strengths and weaknesses of the interface.

What makes me adopt a mixed evaluation approach is that no one single evaluation method can disclose all the usability problems. Heuristic assessment comes in handy in detecting design issues in a cost-effective way, particularly prior to complete development. Task-based usability testing can be applied to monitoring the behaviour of real students in the interface and to determine whether they are able to perform the most important tasks without feeling lost. These techniques combined give theoretical and practical reasons regarding the quality of the prototype.

This assessment plan should be used with Study Mate since the application is designed to be used in daily academic activities. An effective interface should have a clear design, as well as actual user performance. Thus, the strategy is aimed at the review of experts and real interaction with students.

5. Methods Proposed, Motivation and Implementation

5.1 Heuristic Evaluation:

Heuristic evaluation is a form of usability inspection, where evaluators evaluate an interface by comparing it to known usability principles or heuristics. The usability heuristics of Nielsen is one of the most popular collections of heuristics, containing the principles of visibility of system status, user control and freedom, consistency and error prevention. The approach is suitable in the case of Study Mate since the prototype is only at a low-fidelity phase. Heuristic evaluation is economical at this point and comes in handy to detect blatant usability issues prior to further development of the interface. As an example, evaluators can determine the consistency of navigation, a clear system feedback, the ability of users to reverse actions, and avoidance of incomplete input in forms. In order to adopt this approach, the prototype would be assessed by three to five evaluators separately, (Harley, 2018). The individual evaluators would view the screens and match them with the principles of usability. Some of the problems that they would observe include unclear icons, poor feedback, inappropriate labels or bad layout. Following this, there would be an integration of the findings followed by ranking the findings based on severity. The problems of high severity would be given the first priority to be redesigned. The merit of heuristic evaluation is that it can provide a great deal of information about usability problems in a short time, (Nielsen, 1994). It comes in handy particularly

during the initial phases of the design since modifications can be made easily. In the case of Study Mate, this approach will assist in making sure that the prototype already adheres to the fundamental principles of usability prior to its testing by students.

5.2 Task-Based Usability Testing:

Task-based usability testing is the approach whereby the representative users are observed as they struggle to perform real-life tasks with the prototype. This approach targets on user performance, behaviour satisfaction. The evaluator does not simply inquire the users about what they consider but observes what they do, (McCloskey, 2014).

This approach is very appropriate to Study Mate since the application is aimed at the real-life activities. Examples of realistic tasks are:

- add a new assignment, a due date and a reminder,
- finding notes to a given module,
- glance at future deadlines on home page,
- leaving the dashboard and going to the study schedule,
- modifying or deleting a task entry.

The reason behind this approach is that Study Mate needs to be functional not just in theory but also with actual students. Although the interface might be so well-designed, students can experience a problem with labels, workflow or navigation, icons. Task-based testing can be used to demonstrate the ease of use of the app. In order to carry out this approach a small sample of students of various year levels would be chosen. Every participant was to do a series of tasks with the prototype.

The evaluator would keep an eye on the duration of tasks, whether the participant will make mistakes, where the participant will hesitate and whether the participant will be able to accomplish the task during the session.

The participants could also be questioned after the session on short follow up questions on what they found easy or difficult. The outcomes would be thereafter analysed to find out shared usability issues. To use the example, in case some students cannot find the notes section as fast as possible, then the notes icon or label might be enhanced. In case the users are reluctant to save a task, the feedback or button location might require revision. This approach offers first hand results of real usability performance, (Digital gov, 2025).

6. The Conclusion:

The report was an extension of the Study Mate Assignment 1 submission, which enhanced the discussion of the prototype and created a more detailed usability assessment plan. The report indicated how the suggestions in the initial assignment were met with a stronger framework, more elaborated and clearer theory integration. The low-fidelity prototype was designed using five principles, which include visibility, feedback, constraints, mapping and conceptual model. All the principles make the app easier to use, more intuitive and usable by students.

In addition, the study established a practical evaluation plan that combines heuristic evaluation and task-based usability testing. These techniques were chosen because they offer theoretical and practical user insight about the interface's quality. These techniques can be used to further enhance the Study Mate prototype prior to complete development. The goal of developing an academic

productivity app that is practical, easy to use, and in line with student demands is often supported by the design and evaluation process.

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